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Prediction of Static Formation Temperature Considering Real Well Trajectory for Ultra-Deep Well Drilling based on an Efficient Pseudo-3D Ordinary Kriging Model

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Abstract

Accurate prediction of static formation temperature is critical for mitigating drilling risks (e.g., tool failure and fluid degradation) in ultra-deep wells. Existing methods struggle to efficiently predict static formation temperature (SFT) at arbitrary spatial locations while accounting for complex well trajectories. This study proposes an efficient pseudo-3D Ordinary Kriging model integrated with real well trajectories for high-precision SFT prediction. Using data from 112 wells in the Shunbei Oilfield (5.1 million temperature points), 3D trajectories were reconstructed via the minimum curvature method and discretized by true vertical depth. Stratum-specific 2D Ordinary Kriging interpolation was performed on layered horizons, with parameters optimized (θ=[10,10], lob=[0.1,0.1], upb=[20,200]) to minimize RMSE/MAE and maximize R^2. The stacked pseudo-3D model achieved <10% relative error on formation surfaces and <5% along well profiles in deep formations (7500–8500 m). Results demonstrate that incorporating trajectory data eliminates errors from well deviation and captures vertical thermal gradients. Visualization revealed a clear eastward-increasing trend in SFT, correlating with burial depth and formation thickness. The model provides actionable insights for high-temperature drilling design, balancing computational efficiency with geological accuracy.

Introduction

The discovery of increasing volumes of ultra-deep oil and gas resources is driving petroleum drilling towards ultra-deep formations[1, 2]. Ultra-deep formations typically exhibit high static formation temperatures (SFT). Excessively high downhole temperatures during drilling can lead to incidents such as drill bit sticking, alterations in drilling fluid properties, and damage to measurement while drilling (MWD) tools[3]. Therefore, accurately predicting the SFT before drilling ultra-deep wells is critically important.

The primary existing methods for SFT prediction include four categories: direct measurement, drilling fluid temperature recovery inversion, mechanistic modeling, and artificial intelligence (AI) methods. Direct measurement utilizes downhole tools to directly measure SFT. However, this method necessitates a prolonged period of wellbore temperature stabilization[4], a condition often incompatible with the operational timelines of drilling sites. Drilling fluid temperature recovery inversion is the most extensively researched method. It calculates SFT inversely based on the temperature evolution of drilling fluid within the wellbore, transitioning from initial equilibrium with SFT to a cooled state during circulation. This approach includes analytical methods such as the Horner method[5], Leblanc method[6], Manetti method[7], spherical–radial heat flow method[8], and conductive–convective cylindrical source method[9]. A key limitation, however, is that it primarily estimates the SFT in the vicinity of the existing wellbore. Consequently, significant errors can arise if the target well location is distant from the wells used for calculation. Mechanistic modeling involves analyzing heat transfer laws within the formation to compute SFT, employing techniques like proportional–integral control for single-point inversion[10] or establishing numerical heat transfer models[11]. While capable of predicting SFT at arbitrary locations within a region, these models are often complex to construct (typically involving grid discretization and solving inter-grid heat transfer relationships) and generally suffer from slow computational speeds. AI methods typically build upon drilling fluid temperature recovery inversion or mechanistic modeling by incorporating AI algorithms to assist in SFT calculation[12-15]. AI approaches can enhance prediction accuracy and computational efficiency, but training robust AI models usually demands substantial raw data and involves cumbersome training procedures.

Among the conventional SFT prediction methods outlined above, there is a lack of a simple and accurate method for predicting SFT at any arbitrary location within ultra-deep formations for planned wells. Kriging interpolation offers a potential solution to this challenge. Kriging interpolation is a geostatistical technique that predicts attribute values at unsampled locations based on the attribute values and spatial relationships of known data points. Consequently, by utilizing SFT data and well locations from existing drilled wells, Kriging interpolation can predict SFT at any location within the formation. Currently, the Kriging interpolation method is widely applied in geoscience-related fields[16-18], while in drilling and completion engineering, it is more frequently used for geothermal resource assessment[19, 20]. Applying Kriging interpolation to predict SFT in ultra-deep well drilling holds promise for achieving efficient and accurate SFT forecasts.

This study aims to develop a simple and accurate method for predicting SFT at any location within a specific ultra-deep formation. This will be achieved by establishing a Kriging interpolation model that incorporates the influence of the actual wellbore trajectory, utilizing existing SFT data from that formation.

Methodology

This study proposes a trajectory-integrated pseudo-3D Kriging interpolation method for predicting the spatial distribution of SFT. The core methodology comprises two parts: Data Preparation and Preprocessing and Pseudo-3D Kriging Interpolation Model Construction. The specific model construction and application process is illustrated in Fig. 1.

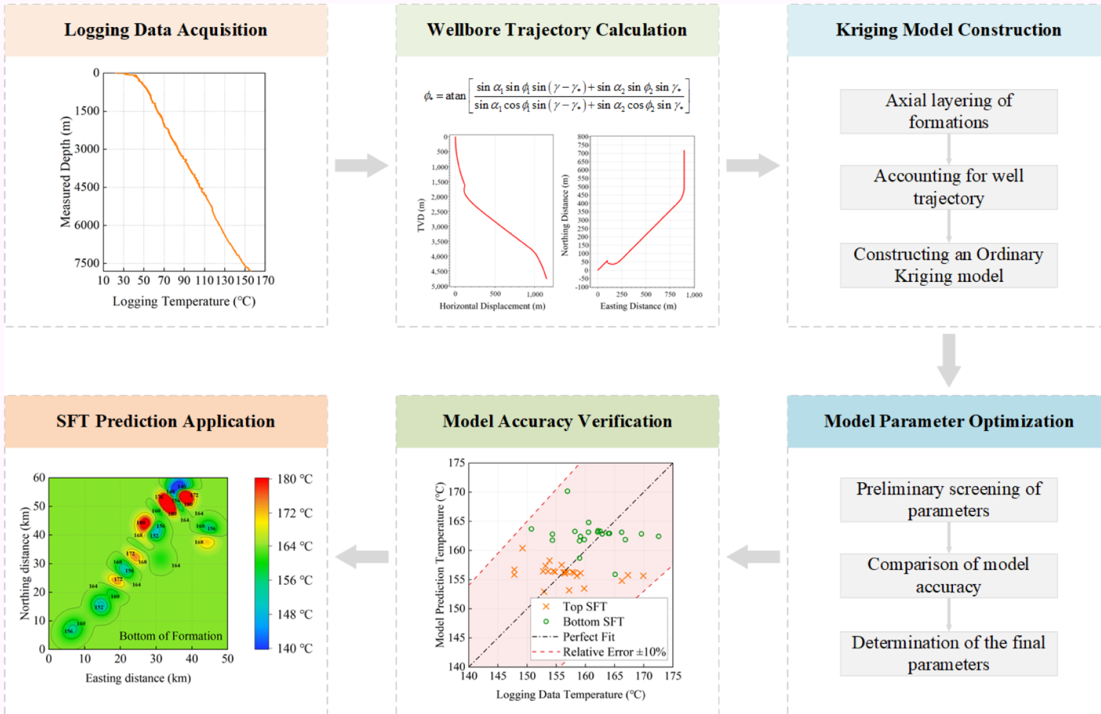

Figure 1—Process for establishing and applying the pseudo-3D kriging interpolation SFT prediction model.

Data Preparation and Preprocessing

The dataset originates from well-logging temperature measurements of completed wells in the Shunbei Oilfield, encompassing 112 wells distributed across an area of approximately 230 km × 170 km. It contains over 5.1 million temperature data points. These wells are primarily directional and horizontal, distributed across multiple fault zones. Measured Depths (MD) predominantly range from 7000 m to 9000 m, with Bottom Hole Temperatures (BHT) generally between 135°C and 200°C. Systematic preprocessing was performed to ensure data representativeness and consistency.

Using well-logging data for each well (including MD, inclination angle, azimuth angle, and wellhead coordinates), the three-dimensional wellbore trajectory was reconstructed using the Minimum Curvature Method (MCM), as shown in Fig. 2.

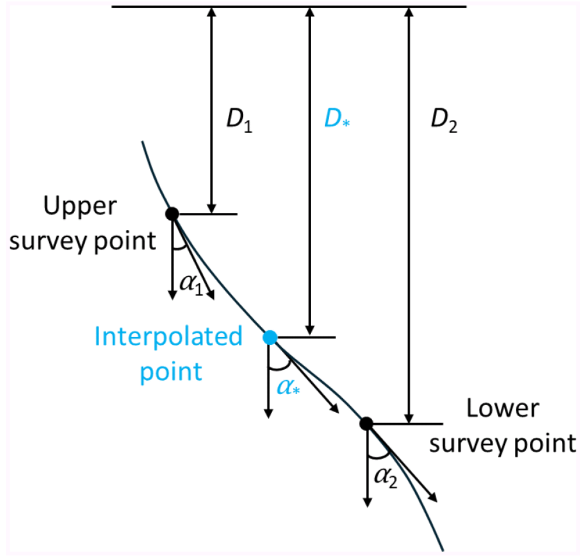

Figure 2—Wellbore trajectory calculation using MCM.

This method calculates the inclination angle and azimuth angle at interpolated points between two known survey points, and subsequently computes the corresponding True Vertical Depth (TVD) and planimetric coordinates. First, the dogleg angle between survey points is calculated:

$$\gamma = \cos^{-1}\left[\cos\alpha_1\cos\alpha_2 + \sin\alpha_1\sin\alpha_2\cos\left(\phi_2 - \phi_1\right)\right] \tag{1}$$

The dogleg angle γ (rad) quantifies the trajectory deviation between adjacent survey stations, calculated using inclination angles (α_1, α_2) and azimuth angles (ϕ_1, ϕ_2) at the upper and lower survey points, respectively.

Subsequently, for a given assumed MD, the dogleg angle relative to the upper survey point is calculated:

$$\gamma_* = \gamma\frac{D_{m,*} - D_{m1}}{D_{m2} - D_{m1}} \tag{2}$$

γ_* (rad) represents the dogleg angle between the interpolated point and upper survey station, with $D_{m,*}$, D_{m1}, and D_{m2} denoting the measured depths (m) of the interpolated point, upper station, and lower station, respectively.

The inclination angle and azimuth angle at the interpolated point are then interpolated using the MCM formulae:

$$\alpha_* = \mathrm{acos}\left(\frac{a\cos\alpha_1 + b\cos\alpha_2}{c}\right) \tag{3}$$

$$\begin{cases} a = \tan\frac{\gamma}{2} - \tan\frac{\gamma_*}{2} \\ b = \tan\frac{\gamma}{2} - \tan\frac{\gamma-\gamma_*}{2} \\ c = \tan\frac{\gamma_*}{2} + \tan\frac{\gamma-\gamma_*}{2} \end{cases} \tag{4}$$

$$\phi_* = \mathrm{atan}\left[\frac{\sin\alpha_1\sin\phi_1\sin\left(\gamma - \gamma_*\right) + \sin\alpha_2\sin\phi_2\sin\gamma_*}{\sin\alpha_1\cos\phi_1\sin\left(\gamma - \gamma_*\right) + \sin\alpha_2\cos\phi_2\sin\gamma_*}\right] \tag{5}$$

α_* and ϕ_* denote the inclination and azimuth angles (rad) at the interpolated point, respectively.

Finally, the TVD and planimetric coordinate increments at the interpolated point are computed:

$$\begin{cases} D_* = D_1 + \Delta D \\ N_* = N_1 + \Delta N \\ E_* = E_1 + \Delta E \end{cases} \tag{6}$$

$$\begin{cases} \Delta D = \lambda_M \left(\cos\alpha_1 + \cos\alpha_* \right) \\ \Delta N = \lambda_M \left(\sin\alpha_1 \cos\phi_1 + \sin\alpha_* \cos\phi_* \right) \\ \Delta E = \lambda_M \left(\sin\alpha_1 \sin\phi_1 + \sin\alpha_* \sin\phi_* \right) \end{cases} \tag{7}$$

$$\lambda_M = \frac{D_{m,*} - D_{m1}}{\gamma_*} \tan\left(\frac{\gamma_*}{2}\right) \tag{8}$$

D_* N_*, and DE_* represent the TVD, northing, and easting coordinates (m) of the interpolated point, respectively; while ΔD, ΔN, and ΔE denote the corresponding TVD, northing, and easting increments (m) from the upper survey station

This process ensures all well temperature data are vertically aligned (based on TVD) for subsequent spatial interpolation.

To precisely map logging temperature data onto the reconstructed 3D well trajectory, each wellbore trajectory was discretized into points at uniform TVD intervals, and the corresponding logging temperature value at each depth was assigned to these trajectory points. During actual logging operations, significant fluctuations can occur in temperature curves due to instrument noise or varying wellbore conditions. These high-frequency fluctuations do not represent true formation temperature variations. Therefore, filtering was applied to the original temperature data to smooth out noise while preserving the underlying trend of temperature change with depth. The filtered temperature data provides a more reliable representation of formation temperature trends, enhancing interpolation model accuracy.

Furthermore, the influence of surface temperature on shallow geothermal gradients was investigated. Analysis of full-borehole continuous temperature logs from several wells revealed that seasonal surface temperature variations primarily affect shallow formations (typically confined to depths above 200–500 m). Beyond 500 m depth, the influence of surface temperature fluctuations becomes negligible. Consequently, for the deep formations of interest in this study, the impact of surface temperature variations on the temperature field can be disregarded, justifying the direct use of logging temperature data to represent the SFT.

Geological formation top data for each well, including depths and thicknesses of formation boundaries, were collected. Based on well completion reports, the top depths of major formations were marked for each well, and the entire well section was subdivided into corresponding stratigraphic units. During the subsequent interpolation modeling, the temperature field was constructed separately for each formation unit to reflect spatial temperature variation characteristics within distinct strata. This stratigraphic control enhances the geological plausibility of the interpolation results: temperature variations within a single formation are governed by its lithology and burial depth, while differences in temperature gradients may exist between formations. Processing by formation layer thus enables a more refined characterization of the temperature distribution within each horizon across the study area.

Workflow for Pseudo-3D Kriging Interpolation Model Construction

When constructing the 3D temperature field model, the study area was first spatially discretized. Vertically, the entire profile was subdivided into multiple formation horizons based on the aforementioned stratigraphic layering. For each formation, beyond the major boundaries along the horizon, several horizontal slices were further defined internally to enhance vertical resolution. To ensure lateral comparability between different formations, the number and size of grid cells on the cross-section of each formation were kept as consistent as possible. This discretization approach accounts for geological layering control while ensuring continuous spatial transition of the temperature field between layers.

For each discretized formation horizon, Ordinary Kriging (OK) interpolation was employed to predict the SFT at grid nodes across the plane. Kriging interpolation is an optimal unbiased estimation method based on the theory of regionalized variables. It leverages the spatial correlation among known sample points to estimate values at unknown locations. Its core principle involves a linearly weighted average of neighboring known data points, with weights determined based on the spatial autocorrelation structure described by a variogram model. For any estimation point s_0 on the plane, its temperature estimate $\hat{Z}(s_o)$ is calculated from surrounding known temperature observations $Z(s_i)$:

$$\hat{Z}(s_0) = \sum_{i=1}^{n} \lambda_i Z(s_i) \tag{9}$$

The weight coefficients λ_i ensure unbiased predictions (zero expected error) under the constraint:

$$\sum_{i=1}^{n} \lambda_i = 1 \tag{10}$$

The weights λ_i are solved using the Kriging system of equations:

$$\begin{bmatrix} C(s_1 - s_1) & \cdots & C(s_1 - s_n) & 1 \\ \vdots & \ddots & \vdots & \vdots \\ C(s_n - s_1) & \cdots & C(s_n - s_n) & 1 \\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \\ \phi_l \end{bmatrix} \begin{bmatrix} C(s_1 - s_0) \\ \vdots \\ C(s_n - s_0) \\ 1 \end{bmatrix} \tag{11}$$

with $C(h)$ denoting the covariance function.

Through the above steps, Kriging interpolation was performed separately for each formation horizon, yielding a 2D temperature distribution map for that horizon.

The 2D interpolation results from each discrete horizon were stacked vertically to form a pseudo-3D temperature field model for the entire study area. The term "pseudo-3D" signifies that the model consists of a series of 2D slices rather than a truly continuous 3D interpolation. This approach achieves a balance between computational efficiency and accuracy: On one hand, interpolation by layer ensures the accuracy of the temperature field within each layer and captures differences in temperature gradients between different formations. On the other hand, after merging results vertically, the model exhibits continuous temperature distribution longitudinally, allowing temperature queries at any arbitrary point. The final constructed 3D temperature field model is stored in grid format, with each grid node containing its coordinates and corresponding predicted temperature value. Visualization of this model enables intuitive representations of formation temperature variations with spatial location and depth within the study area, such as isotherm maps on different depth planes and temperature distribution profiles along specific sections.

Results & Discussion

The original data used in this study were primarily obtained from the five fault zones in the Shunbei Oilfield, designated as Fault #1, Fault #2, Fault #3, Fault #4, and Fault #5. The well-logging temperature data are notable for their large quantity and high precision, which are critical for reliable model predictions. Therefore, selecting a fault zone with abundant and high-quality well-logging temperature data as the data source will make the analysis of model prediction results more effective. Considering both the quantity and quality of the well-logging temperature data from each fault zone, Fault #1 was selected as the primary dataset for model validation and result analysis. Fig. 3 shows the distribution of formation positions recorded by well-logging data in Fault #1 of the Shunbei Oilfield extends from west to east. To enhance model accuracy through parameter adjustment and validate its precision, this study utilizes the Yijianfang Formation of the fault as the source of the original model data. This formation is primarily distributed at depths ranging from 7500 m to 8500 m below ground surface, deepening gradually from west to east with an increasing formation thickness. This formation was selected as the data source for model validation for the following primary reasons: (1) The formation depth is significant, resulting in high formation temperatures typically exceeding 140 °C, enabling the validation of model performance under high-temperature formation conditions; (2) The formation is continuous and contains a substantial number of original data points, which is conducive to achieving better model prediction results; (3) According to well history records, this formation contains hydrocarbon-bearing reservoirs, making SFT prediction for this formation highly relevant to practical oil and gas well drilling operations. This section will analyze the impact of several adjustable parameters within the model on model accuracy, optimize the set of parameters yielding the highest accuracy, validate the model's calculated accuracy against measured data, and finally present visualizations demonstrating the model's practical application effectiveness.

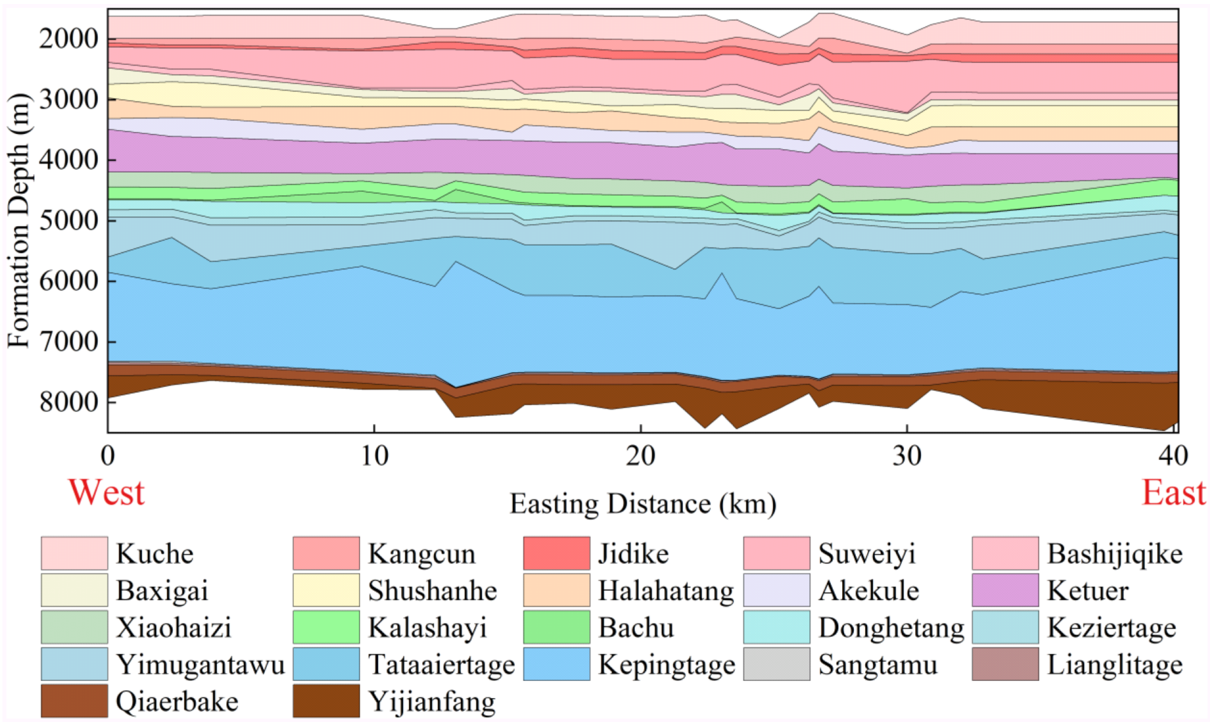

Figure 3—Formation distribution in Fault #1 of the Shunbei Oilfield.

Parameter Optimization

The model incorporates three types of parameters: θ, lob, and upb. Each type consists of two adjustable parameters (θ_1, θ_2; lob_1, lob_2; upb_1, upb_2). Each parameter controls the influence of its parameter type on the model in the horizontal and vertical directions, respectively. θ is a correlation length parameter that governs the spatial correlation range. It defines the distance over which known data points influence prediction points in both the horizontal and vertical directions. A larger θ indicates a broader spatial correlation range but may reduce the overall model accuracy. Conversely, a smaller θ increases sensitivity to local variations, potentially improving accuracy near data points but deteriorating prediction accuracy at unsampled locations farther away by weakening their relationship with the known points. lob represents the lower bound constraint, specifying the minimum allowable value for the correlation length θ during the optimization process. It prevents θ from becoming excessively small, thereby avoiding overfitting and ensuring the model does not overemphasize local data variations. upb represents the upper bound constraint, defining the maximum allowable value for θ during optimization. It prevents θ from becoming too large, mitigating underfitting and preventing the model from becoming oversmoothed and ignoring local details. During model construction, initial screening of these three parameter types was performed to ensure basic model accuracy.

Parameter optimization was conducted by computing SFT using different parameter sets and comparing the results with the original logging temperature data. The comparison between model predictions and measured logging temperatures was evaluated using three metrics: root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2). The parameter set yielding high and stable accuracy across all three metrics was identified as the optimal configuration. Fig. 4 presents the parameter optimization process and results for a representative well.

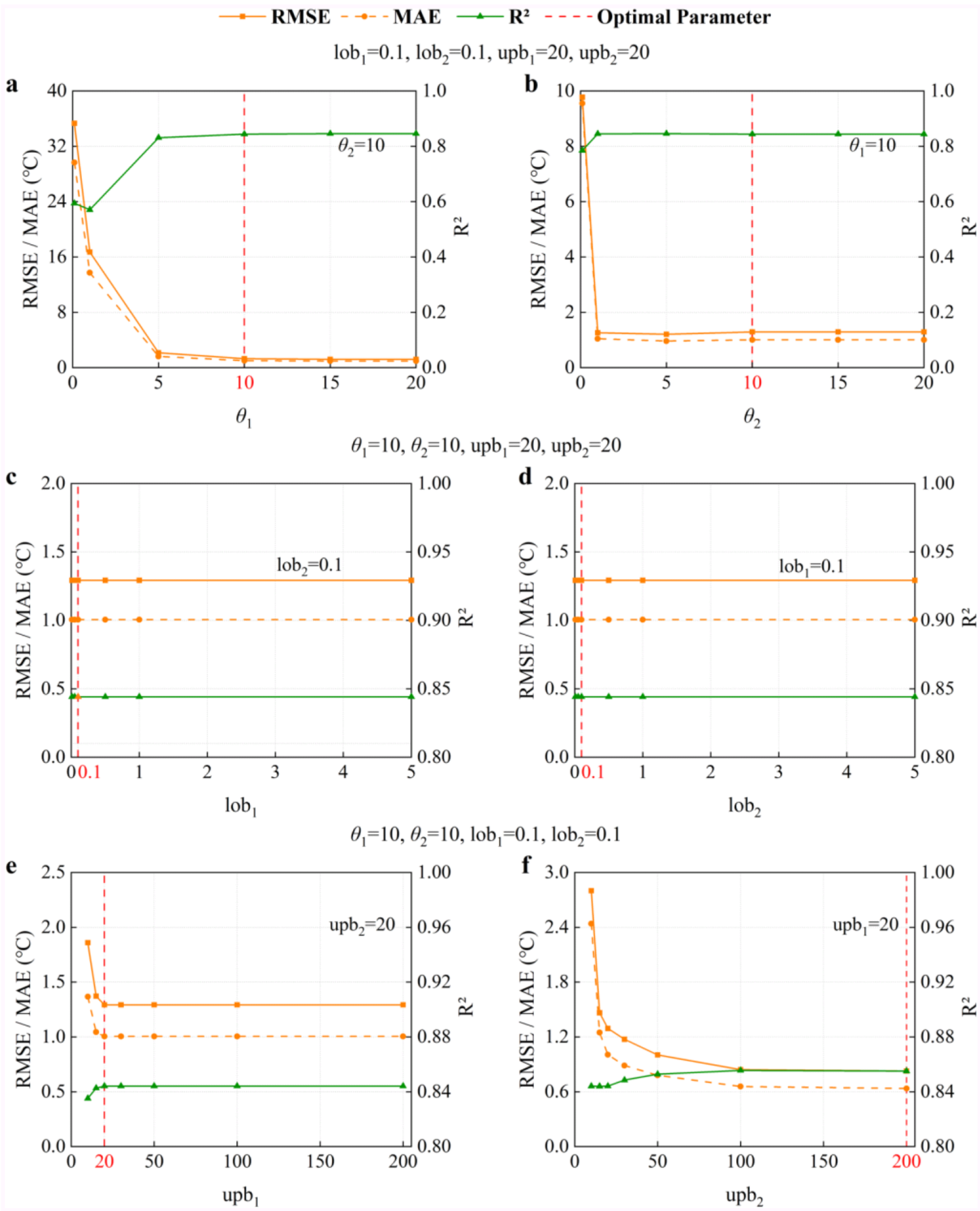

Figure 4—Parameter optimization of the pseudo-3D Kriging model: (a) Optimization of θ_1; (b) Optimization of θ_2; (c) Optimization of lob_1; (d) Optimization of lob_2; (e) Optimization of upb_1; (f) Optimization of upb_2.

For θ optimization, six values were tested: 0.1, 1, 5, 10, 15, and 20. As shown in Fig. 4(a) and Fig. 4(b), for both θ_1 and θ_2, RMSE and MAE gradually decreased while R^2 increased as the θ value increased. This indicates that model accuracy progressively improved and stabilized with larger θ values. When both θ_1 and θ_2 exceeded 10, the model achieved its highest accuracy. Further increases in θ did not alter RMSE, MAE, or R^2, signifying that model accuracy plateaued. Based on this analysis of accuracy trends, θ_1=10 and θ_2=10 were selected as the optimal values for θ.

For lob optimization, five values were tested: 0.01, 0.1, 0.5, 1, and 5. As shown in Fig. 4(c) and Fig. 4(d), RMSE, MAE, and R^2 remained unchanged regardless of the lob value, indicating no variation in model accuracy. The lob parameter prevents the model from being overly influenced by local SFT variations. However, in this study, the significant distance between the wells providing the original data ensured the absence of excessive local SFT fluctuations. Consequently, changes in lob did not impact model accuracy. Therefore, lob_1=0.1 and lob_2=0.1 were selected as the optimal values for lob.

For upb optimization, seven values were tested: 10, 15, 20, 30, 50, 100, and 200. As shown in Fig. 4(e) and Fig. 4(f), for both upb_1 and upb_2, RMSE and MAE gradually decreased while R^2 increased as the upb value increased, reflecting improved model accuracy. However, the magnitude of this improvement differed between the parameters. Model accuracy reached its peak at upb_1=20, with no further improvement observed for higher values. In contrast, model accuracy continued to increase with larger upb_2 values, albeit at a diminishing rate. The model achieved high accuracy at upb_2=200. Given that 200 represents a relatively high value for upb, higher values were not analyzed. Consequently, upb_1=20 and upb_2=200 were selected as the optimal values for upb.

In summary, the model achieved its highest accuracy with the parameter configuration θ=[10, 10], lob=[0.1, 0.1], upb=[20, 200]. This parameter set will be employed in all subsequent research involving this model.

Model Accuracy Validation

This study evaluates the accuracy of the model predictions through two analytical dimensions: precision assessment in the radial distribution direction of the formation, and precision assessment along the wellbore trajectory profile. This comprehensive approach validates the model's spatial accuracy across the entire domain.

Fig. 5 compares the model-predicted SFT with measured SFT values at the top and bottom surfaces of the Yijianfang Formation. The data presented were generated by treating one drilled well within the fault as the prediction target, utilizing SFT data from all other wells in the fault as input for the model, and then comparing the predictions against the measured SFT. The predicted SFT values for most well locations in the Yijianfang Formation closely align with the measured SFT data. The red-shaded areas denote predictions with relative errors below 10%, an accuracy level achieved by all data points. This demonstrates the model's high accuracy on the planar surface along the formation distribution.

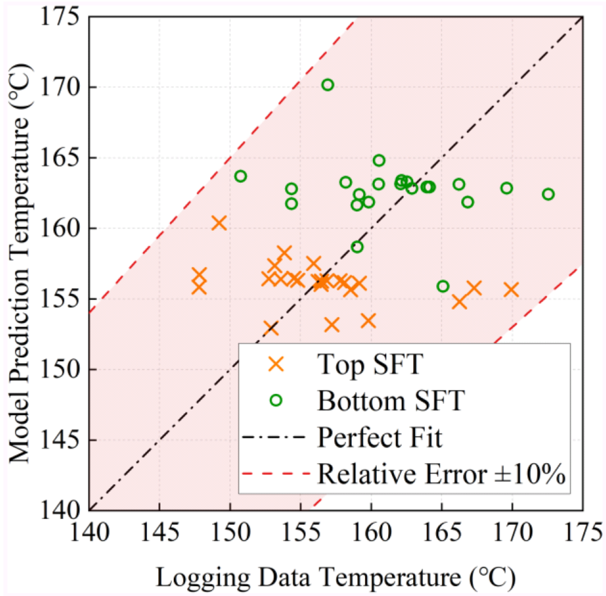

Figure 5—Model accuracy validation against measured SFT data on the planar surface of the Yijianfang formation.

To visually illustrate the model's precision across the planar surface of the Yijianfang Formation, the predicted SFT values at the top surface for all modeled locations were compared against measured SFT data, ordered from west to east, as shown in Fig. 6. The measured SFT at the top surface predominantly clusters around 155 °C, a trend successfully captured by the model predictions. Furthermore, all model predictions exhibited relative errors below 10%, with the majority under 5%. This further confirms the model's high accuracy on the planar surface along the formation distribution.

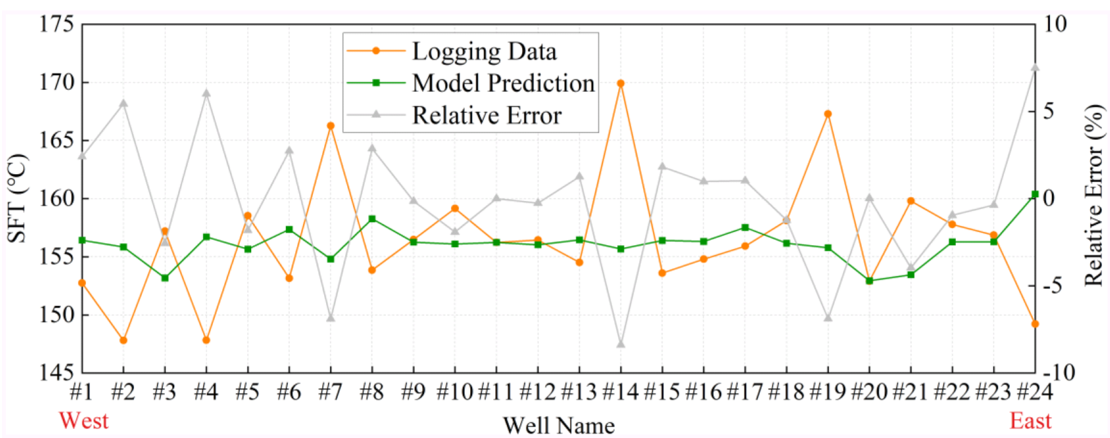

Figure 6—Modeled vs. measured SFT at the top surface of the Yijianfang formation.

Fig. 7 compares the model-predicted axial SFT profile (along the wellbore) within the Yijianfang Formation against measured SFT data. The model predictions used measured SFT data from other wells within the fault as input. The figure presents comparisons for six wells, showing that all predictions achieved relative errors below 5%. This indicates the model's high accuracy in axial SFT prediction. Additionally, the high parallelism between the predicted and measured SFT profile curves is notable. Since the slope of these curves represents the geothermal gradient, the model-predicted SFT data can also accurately compute geothermal gradients.

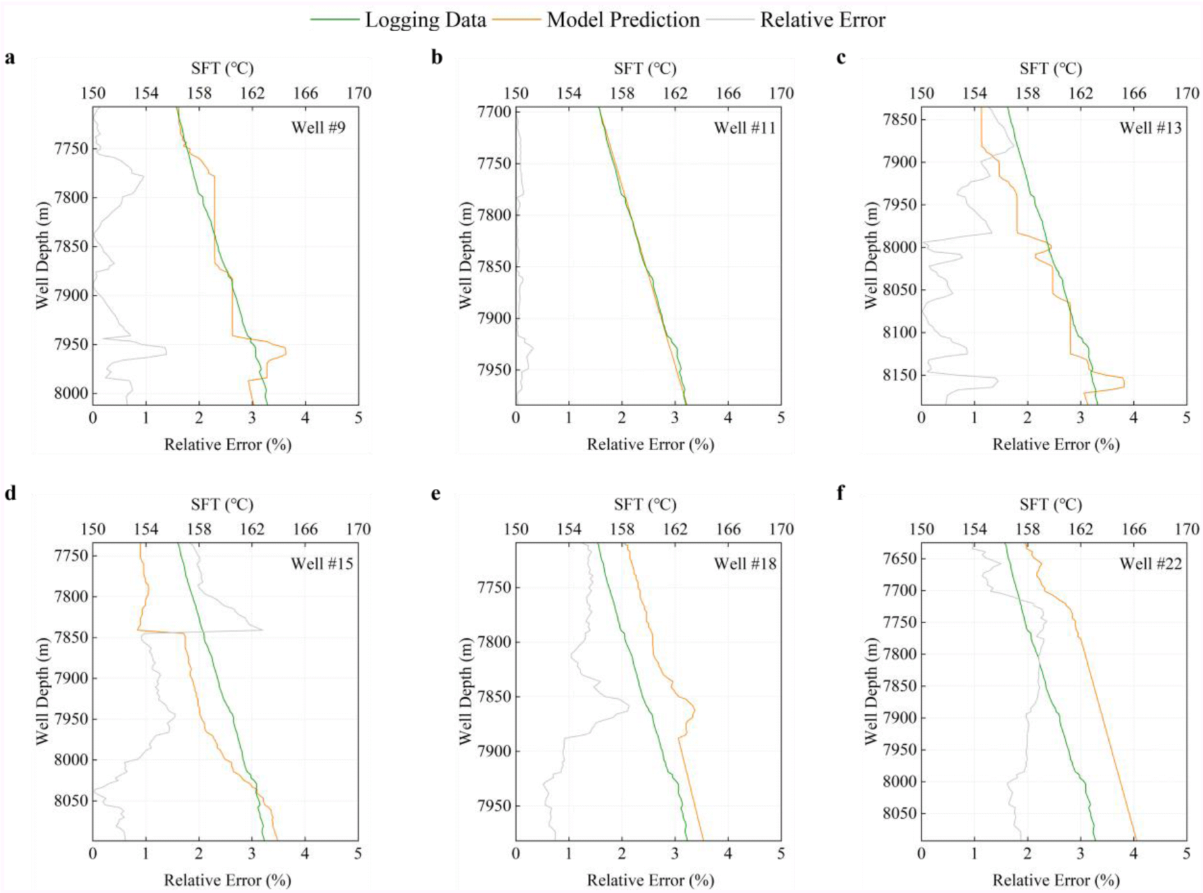

Figure 7—Model accuracy validation against measured SFT data on the formation profile of the Yijianfang formation: (a) Validation of Well #9; (b) Validation of Well #11; (c) Validation of Well #13; (d) Validation of Well #15; (e) Validation of Well #18; (f) Validation of Well #22.

In summary, the model demonstrates high accuracy in SFT prediction, both on the planar surface along the formation and along the axial wellbore profile within the formation. Consequently, the model can accurately compute SFT at any arbitrary point within a target formation across a given region. The model's high accuracy endows it with significant practical value for field applications.

SFT Temperature Field Visualization

The established high-accuracy SFT prediction model enables visualization of the SFT of specific formations and subsequent analysis of its spatial distribution patterns. These analyses provide critical insights to better guide high-temperature drilling operations in ultra-deep wells. Fig. 8 shows the SFT contour map of selected deep formations within Fault #1.

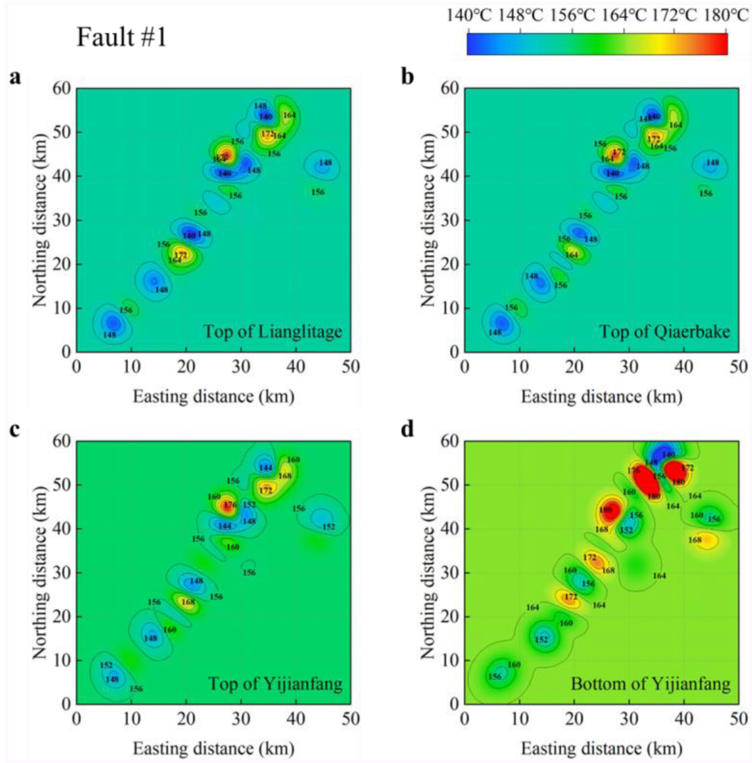

Figure 8—SFT distribution contour map of Fault #1: (a) SFT distribution at the top of the Lianglitage formation; (b) SFT distribution at the top of the Qiaerbake formation; (c) SFT distribution at the top of the Yijianfang formation; (d) SFT distribution at the bottom of the Yijianfang formation.

As shown in Fig. 8, the SFT distribution across the planar surface of this fault exhibits a distinct trend: temperatures increase progressively from west to east and from south to north. This spatial pattern aligns with the SFT distribution observed in Fig. 6. Furthermore, by comparing the SFT distributions across formations at different depths, the distribution pattern of SFT along the stratigraphic profile can be investigated. As shown in Fig. 8, the SFT of the deeper formation is significantly higher than that of the shallower formation, which is consistent with the characteristic increase of SFT with greater formation depth. Regarding the SFT distribution at the top and bottom of the Yijianfang Formation, examination of the specific SFT values on these two surfaces reveals that the difference between the bottom SFT and top SFT is smaller in the northeastern than in the southwestern. With reference to Fig. 3, it can be observed that the Yijianfang Formation in this fault zone is thinner in the west and thicker in the east, with the eastern section also buried deeper. As a result, the difference between the bottom and top SFT within the same formation varies spatially.

To further demonstrate the application effectiveness of the established model in this study, well-logging temperature data from deep formations in other fault zones of the Shunbei Oilfield were input into the model to predict the SFT of the corresponding formations, and contour maps were generated, as shown in Figs. 9 to 12. These SFT distribution contour maps allow for intuitive visualization of the planar SFT distribution patterns of a specific formation within a given fault zone.

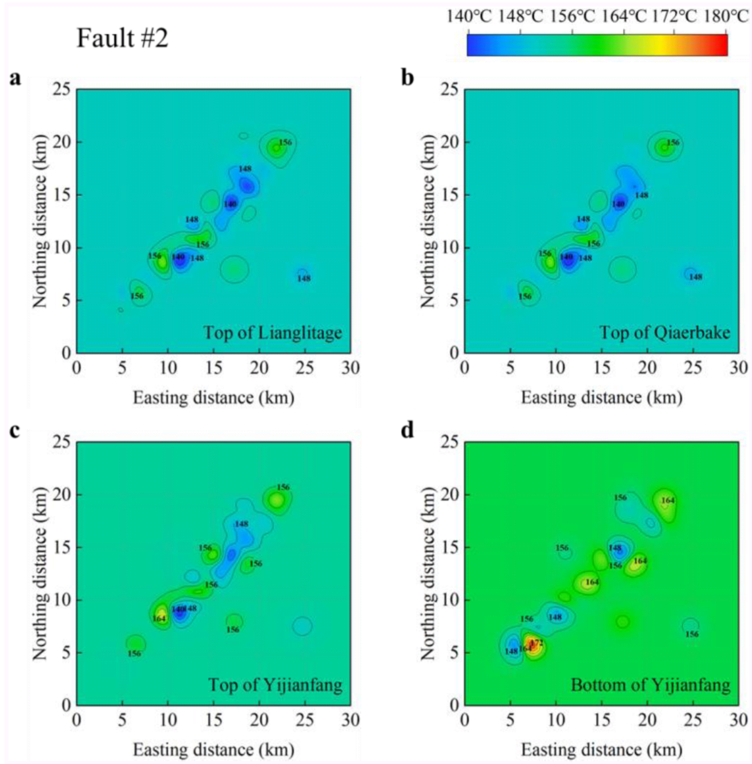

Figure 9—SFT distribution contour map of Fault #2: (a) SFT distribution at the top of the Lianglitage formation; (b) SFT distribution at the top of the Qiaerbake formation; (c) SFT distribution at the top of the Yijianfang formation; (d) SFT distribution at the bottom of the Yijianfang formation.

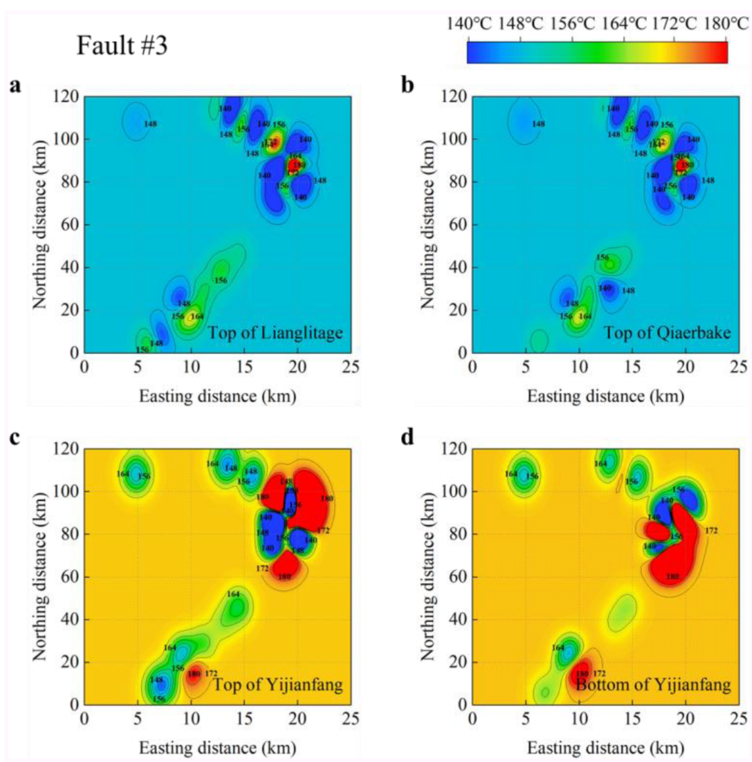

Figure 10—SFT distribution contour map of Fault #3: (a) SFT distribution at the top of the Lianglitage formation; (b) SFT distribution at the top of the Qiaerbake formation; (c) SFT distribution at the top of the Yijianfang formation; (d) SFT distribution at the bottom of the Yijianfang formation.

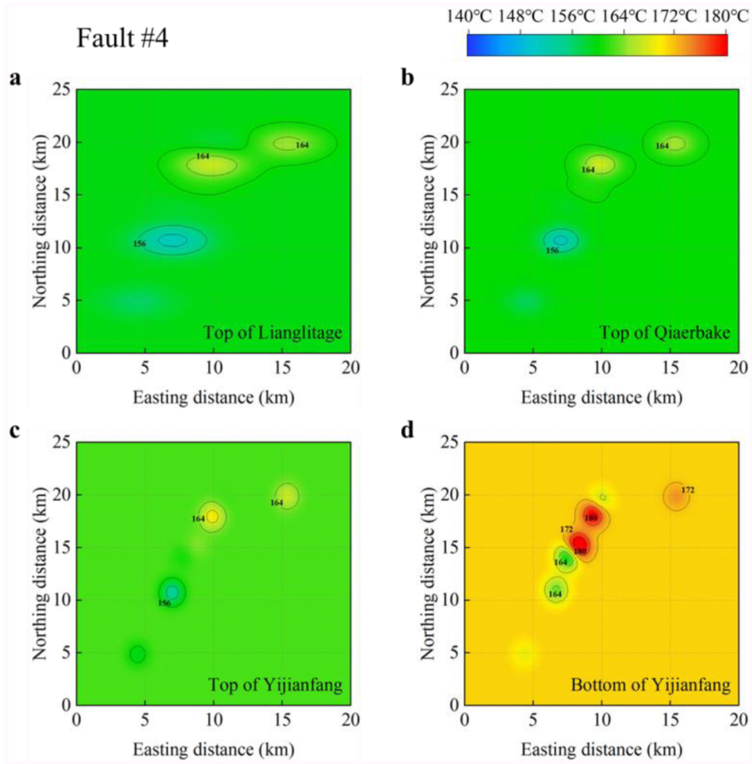

Figure 11—SFT distribution contour map of Fault #4: (a) SFT distribution at the top of the Lianglitage formation; (b) SFT distribution at the top of the Qiaerbake formation; (c) SFT distribution at the top of the Yijianfang formation; (d) SFT distribution at the bottom of the Yijianfang formation.

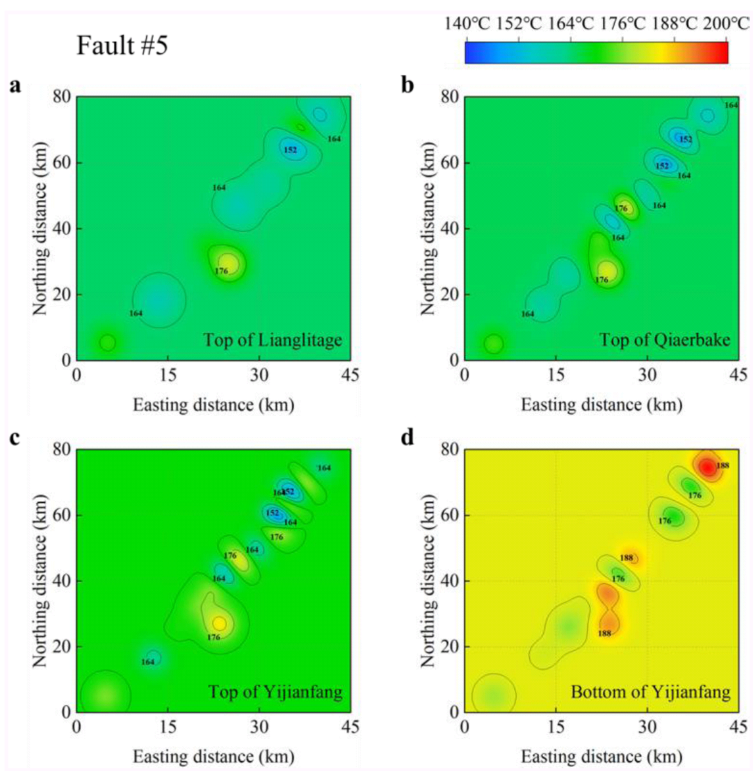

Figure 12—SFT distribution contour map of Fault #5: (a) SFT distribution at the top of the Lianglitage formation; (b) SFT distribution at the top of the Qiaerbake formation; (c) SFT distribution at the top of the Yijianfang formation; (d) SFT distribution at the bottom of the Yijianfang formation.

Conclusions

This study establishes an efficient pseudo-3D Ordinary Kriging model integrating real well trajectories for predicting SFT in ultra-deep drilling environments. Based on extensive validation using field data from the Shunbei Oilfield, the following conclusions are drawn:

1. The proposed pseudo-3D Kriging model achieves high-precision SFT prediction at arbitrary spatial locations within ultra-deep formations. By leveraging existing well-logging temperature data and reconstructing 3D well trajectories via the MCM, the model generates a stratigraphically constrained temperature field. Validation against measured data from the Yijianfang Formation (7500–8500 m depth) demonstrates relative errors below 10% on formation surfaces and under 5% along wellbore profiles, confirming its robustness in high-temperature environments.
2. Integration of actual well trajectories is critical for capturing true spatial temperature heterogeneity. Traditional Kriging methods often overlook wellbore deviations, leading to significant errors in directional and horizontal wells. This study resolves this by aligning temperature data with TVD through trajectory discretization and MCM-based coordinate calculations. The resulting pseudo-3D framework ensures geological plausibility by preserving layer-specific thermal gradients while enabling continuous vertical temperature queries.
3. The model optimizes computational efficiency without sacrificing accuracy through a layered pseudo-3D approach. By discretizing formations into horizons and performing 2D Kriging interpolation per layer before vertical stacking, the method avoids the complexity of full 3D gridding. Parameter optimization (θ=[10, 10], lob=[0.1, 0.1], upb=[20, 200]) further enhances performance, reducing RMSE and MAE while raising R^2. This balances the need for regional-scale applicability with the resolution required for drilling engineering decisions.
4. Visualization of the 3D SFT field reveals actionable insights for ultra-deep drilling operations. Contour maps of Fault #1 exhibit a clear west-to-east temperature increase, correlating with burial depth and formation thickness variations. The quantified vertical thermal gradients provide direct input for drilling fluid design and tool deployment, mitigating risks such as fluid degradation and MWD tool failure in high-temperature zones.

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